Tamanjyot Singh

The University of Texas at Dallas, Richardson, TX, USA

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Bio

Aerospace engineer pursuing a **doctorate in physics** with specialization in astrophysics. Interested in understanding **black hole spin** dynamics and their implications. Currently working as a research assistant at the University of Texas at Dallas under the supervision of Dr. Michael Kesden. My thesis work focuses on (i) inclination distributions in **tidal disruption events** by spinning supermassive black holes and their observational implications, (ii) **precessional and nutational** effects on the gravitational wave signals originating from inspiralling stellar mass binary black hole as currently observed by LIGO, and (iii) **degeneracies between lensed and precessing** gravitational wave signals for current LIGO sensitivities.

Education

Ph.D. — **The University of Texas at Dallas** — Richardson, TX, USA
Physics — Jan 2021 - May 2026 *(expected)*
Thesis: *The Black Hole Ballet*: Spin and precession effects on gravitational waves and tidal disruption Events
Advisor: Dr. Michael Kesden

M.S. — **The University of Texas at Dallas** — Richardson, TX, USA
Physics — Jan 2021 - Dec 2023
GPA: 3.93/4.0

B.Tech. — **Punjab Engineering College (Deemed to be University)** — Chandigarh, India
Aerospace Engineering with minor specialization in Mathematics — Aug 2016 - Jun 2020
GPA: 9.2/10.0, Rank 2
Major project: Computational study of flame propagation in a micro-scale channel and applications of micro-combustors in thermo-electric power generation.

Experience

Research assistant — Richardson, TX, USA
The University of Texas at Dallas — May 2022 - Ongoing

- **Lensing and Precession**: Distinguishing between a lensed and precessing gravitational wave signal, exploring degeneracies and establishing differences.
- **Precession in gravitational waves**: Investigating the effects of spin dynamics in a binary black hole system on the gravitational wave emission.
- **Tidal disruption events**: Solving for inclination biases induced by spin breaking the spherical symmetry in stellar disruptions by supermassive black holes.

Responsibilities: Meeting weekly, developing a code, working in a team environment, mentoring students, and preparing manuscripts for publication.

Teaching assistant — Richardson, TX, USA
The University of Texas at Dallas — Jan 2021 - May 2022

- **Physics Laboratory I: Mechanics**: Introductory physics laboratory course for all STEM undergrad tracks.

Responsibilities : Hosting weekly sessions for students to guide them through the mechanics lab experiments. Grading their lab report submissions.

Research Intern — Mumbai, India
Combustion research Laboratory, IIT Bombay — Jan 2019 - Apr 2019

- **Combustion characteristics and effect of hydrogen content on syngas-air mixtures**: Formulated an analytical correlation for laminar burning velocity of multi-component gaseous mixtures simultaneously at elevated pressures and temperatures.

Responsibilities: Preparing weekly progress updates for group meetings. Writing a report and presenting research at Punjab Engineering College, Chandigarh at the end of program.

IAF (Indian Air Force) Intern — Chandigarh, India
3 BRD, Air Force Station — June 2018

Publications

Published:

"Distribution of orbital inclinations for tidal disruption events by Kerr black holes" — February 2024
Tamanjyot Singh, Michael Kesden
10.1103/PhysRevD.109.043016

Talks

Contributed	**Team for EXtragalactic AStrophysics (TEXAS) – Partnerships in Astronomy & Astrophysics Research and Education (PAARE) Annual Symposium 2024**, The University of Texas at Dallas, Richardson, TX *Title*: "Gravitational wave signals from precessing and nutating binary black holes"	April 2025
Contributed	**APS Global Physics Summit 2025**, Anaheim, CA *Title*: "Gravitational wave signals from precessing and nutating binary black hole inspirals"	March 2025
Contributed	**Fall meeting 2024, Texas Section APS**, Southern Methodist University, Dallas, TX *Title*: "Signatures of generic precession in gravitational wave signals from merging binary black holes"	October 2024
Invited Visit	Center for Computational Astrophysics, Flatiron Institute, New York City, NY *Title*: "Precession in GWs"	May 2024
Contributed	**APS April meeting 2024**, Sacramento, CA *Title*: "Gravitational wave signals from precessing binary black hole inspirals"	April 2024
Contributed	**Team for EXtragalactic AStrophysics (TEXAS) – Partnerships in Astronomy & Astrophysics Research and Education (PAARE) Annual Symposium 2024**, University of North Texas, Denton, TX *Title*: "Tidal disruption events: Inclination distributions for spinning supermassive black holes"	March 2024
Contributed	**10th Gulf Coast Gravity Meeting 2024**, University of Texas, Austin, TX *Title*: "Gravitational wave signals from precessing binary black hole inspirals"	March 2024
Contributed	**International Conference on Gravitation and Cosmology 2023**, Indian Institute of Technology, Guwahati, Assam, India *Title*: "Inclination distributions in tidal disruption events by spinning supermassive black holes"	December 2023
Contributed	**Fall meeting 2023, Texas Section APS**, Angelo State University, San Angelo, TX *Title*: "Identifying Precession in Gravitational Wave signals from Binary Black Hole inspirals"	October 2023
Contributed	**Spring meeting 2023, Texas Section APS**, Texas A&M University, Commerce, TX *Title*: "Tidal disruption events by spinning supermassive black holes" (see page 19)	March 2023

Achievements

NSM Conference Travel Award by School of Natural Sciences and Mathematics at The University of Texas at Dallas, TX	March 2025
APS April meeting DGRAV Travel Award	April 2024
Margie Renfrow travel award by the Department of Physics, The University of Texas at Dallas, TX	March 2024
Teaching Assistant award 2022 at The University of Texas at Dallas, TX	December 2022
PECOSA Sydney Prize for year 2017-18 (Merit Based - Awarded to 3 out of 700+ students)	September 2018
1st position in the Mathematics Quiz sponsored by Punjab State Council for Science and Technology, Chandigarh (Gold Medal - District Level Competition)	September 2016
99.3%ile in JEE (Mains) 2016 (among 1,500,000+ candidates all over India)	April 2016
International Rank 76 in International Olympiad of Science 2013 (Gold Medal + *Special Achievement Certificate*)	November 2013

Workshops and Course Certificates

Attended **Entering Mentoring** at the University of Texas at Dallas	July 2024
Course Certificate for **Astro 101 : Black Holes** by University of Alberta on Coursera.	June 2020
Hosted a workshop at undergrad level for **Gravitational lensing** at Punjab Engineering College, Chandigarh	October 2019
Attended **ASTROWIN**, Winter School on Astronomy 2019 at the Birla Museum, Hyderabad	February 2019
Course Certificate for **Particle Physics: an Introduction** by University of Geneva on Coursera.	July 2018
Course Certificate for **Understanding Einstein: The Special Theory of Relativity** by Stanford University on Coursera.	July 2017

Mentoring

I have had a chance to be a secondary mentor to many undergraduate and near-peer mentees over the years. Following is a list of names and roles of mentees:

Tien Nguyen	Partnerships in Astronomy & Astrophysics Research and Education (PAARE) fellow 2023-24 *(Post-baccalaureate student)*
Maya Basu	Summer REU student 2024 *(Undergrad Student)*
Tinatin Tsiskaridze	Summer exchange student 2024 *(Grad Student)* (from Tblisi State University, Georgia)
Niel Shah	Summer REU student 2023 *(Undergrad Student)*
Deven Linthicum	Summer REU student 2023 *(Undergrad Student)*
Allison Powell	Summer REU student 2022 *(Undergrad Student)*

Extra Curricular Responsibilities

Journal Club host: Leading and hosting weekly meetings with astrophysics students and faculty across DFW in a joint journal club with University of North Texas (UNT), Denton and University of Texas (UT) Arlington. Graduate students encouraged to present a recent paper in their field of research (of their choice) at least once per semester.	August 2024 - Ongoing
Graduate Students in Physics (GSP) Vice President: Representing graduate student interests in	May 2023 -

the department. Responsible for assisting President in maintenance and workings of the organization. *Standout*: Procured a department-wide Overleaf license. July 2024

Graduate Students in Physics (GSP) Astronomy committee head: Grad student liaison for all things telescope, hosting star-watch and lunar eclipse parties. Responsible for maintenance of the existing equipment and inventory along with buying new equipment. March 2022 - February 2024